Instructor key: Paper 009

Use after attempting the quiz. Equivalent scoped answers count. The local rubric awards result, reason, and scope for each question. **AuthorityEffect: None.** This key provides practice feedback, not a credential.

Level 1

1. Example: on the declared triangle, check whether replacement (5,0,0) of original (1,1,1) is refused as `DefectIncrease` and active claims remain unchanged. Name the native-backed checker, score definition, revision, command, and stored state. This is a proposed check until executed.
2. Assumption: declared graph and allowed potential move. Witness: proposed potential. Receipt: observed execution record. Obligation: missing native replay. All except an observed receipt may precede execution.
3. `qed` is an assertion; a matching hash identifies declared bytes relative to the digest. The theorem needs proof/check under stated hypotheses. Custody does not authenticate the producer or establish truth.

Level 2

4. df=(1/3,1/3,−2/3), h−df=(0,0,1). Original and residual sums are both 1.
5. (f1−f0)+(f2−f1)+(f0−f2)=0. Constant shifts cancel on every edge. This argument covers all potentials on the declared cycle.
6. df=(2,−1,−1), h−df=(0,0,5), original/residual sum 5. Replacement by df has sum 0 and changes input; it does not remove the original sum.

Level 3

7. Both witnesses must be feasible. Positive: feasible matching witnesses. Separate negatives: violate a primal bound; introduce nonzero dual vertex flow; retain feasibility but choose a different objective. Proposed or source-inspected controls are not fresh results until executed.
8. Recomputed digest from changed bytes should fail comparison with the old key. String length checks only shape; neither test establishes content truth.
9. No: always-pass accepts the valid case too. Pair a valid bound node with mutated content retaining its old key. Expect pass/failure respectively. The distinction is byte binding under declared serialization.

Level 4

10. TP=6, FN=0, FP=2, TN=2, with the positive conventions in the question. Refused row 6, not-refused row 4, total 10.
11. Challenged/refused=6/6=1; challenged/not-refused=2/4=1/2; refused/challenged=6/8=3/4; label agreement=(6+2)/10=4/5. These

describe these records under these historical labels.

12. The caught flag is also true for both concurrences. The retained later checker replay refuses all ten; the four obstruction claims lack checked witnesses. Earlier "sound" meant only not refused at that time. No physical ground truth or general rate follows. No adversary blocks anything the checker would otherwise accept. The later receipt does not pin its checker revision, and reading it does not freshly replay it.

Level 5

13. The named report lists 32 theorems (28 axiom-free, four with dependencies); the full report includes 516 module declarations, including generated and private ones. Inspection does not compile Lean. Fresh compilation must record the pinned toolchain and actual commands.
14. Equal self-hashes fail an unequal-hash predicate. Distinct hashes do not prove statistical independence or independent error sources. Runtime correspondence is a separate obligation.
15. Scores need independent computation from original inputs; flags need provenance and runtime correspondence. Supplied-value routing is a narrow fixture test, not comprehensive semantic verification.

Level 6

16. Act IV replaces the cochain with zeros; Act II scores those zeros. The original sum 3 cannot be removed by potential subtraction. This is a constructed failure at a historical commit, not its measured frequency.
17. F2a misses a counterexample without keywords; F2b blocks a negated keyword. The pair isolates lexical adjudication. Five probes, four defect families.
18. F3's run-key hash does not identify stored evidence, and constant flags do not check balance or norm bounds. F4's chain binds unsupported template text, including unexecuted repairs. Consistent bytes do not verify claims.

Level 7

19. Necessary conditions: eligible candidate, no applicable blockers, and an accepted checker receipt bound to the candidate. A declared carrier also requires verified exact joint zeros. With no declared carrier, that zero requirement does not apply. A supplied zero alone is insufficient. This describes the pinned correction's source, not a fresh gate execution.
20. Binding identifies bytes; authentication identifies a producer; correctness establishes the checker implements the claimed property; critique validity requires a checked argument. The latter do not follow from digest equality or successful typed parsing.
21. "I inspected the pinned correction and read the historical reports of 66 and 44 passing tests; I did not execute them." A fresh claim needs the checkout,

executable/dependency pins, named test commands, dated outcomes, and preserved outputs. Historical JSON inspection is a different operation.

Level 8

22. (a) Supported within scope; (b) Refused claim; (c) Open obligation; (b) Not run. Different claims can have different dispositions.
23. Claim, revision, path/digest, assumptions, exact input, command, environment or executable pin, result, disposition, obligation. RFC 8785 also requires primitive serialization and other conditions. A zero-denominator fraction is undefined, represented by `None`, not zero or perfect performance.
24. Record unchanged residual, attempted move family, and invariant. New representations need new assumptions and checks. A trigger does not license protected input changes or demonstrate an effective search escape.

Capstone key (8 points)

1. Three retained digests match provenance. This establishes custody relative to the declarations, not independent authentication of the execution.
2. Exactly ten distinct conditions: two named lattices, values $-4,-2,0,2,4$. The row count excludes duplicates hidden by a set.
3. Cells (6,0,2,2); fractions 1,1/2,3/4,4/5 with Question 11 meanings.
4. One refused challenge changed to concurrence gives (5,1,2,2), and 5/6 in the refused row. The copied control is not a new model trial.
5. Changing only the caught flag leaves cells unchanged.
6. F1 input replacement; F2a/F2b keyword adjudication; F3 phantom references; F4 templated receipts. Five records, four families.
7. Preserve original files and identify every altered copy as a control.
8. Example: "At the provenance-bound inputs, `python tools/review_capstone.py` passed custody, census, and recount controls, yielding (6,0,2,2). This freshly recounts retained data; no model, native checker, or gate was rerun. A native replay still needs the checker revision and executable." Add your actual environment, date, and artifact identities. The tool does not grade prose.